

# Mathematical Foundations of Computer Graphics & Vision



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Based on slides by Karlis Briedis

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# Today!

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- Handout, exercise 3
- Q&A

# Info

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## Next week

- Deadline for Exercise 2: 26.03.2024 at 11:59pm

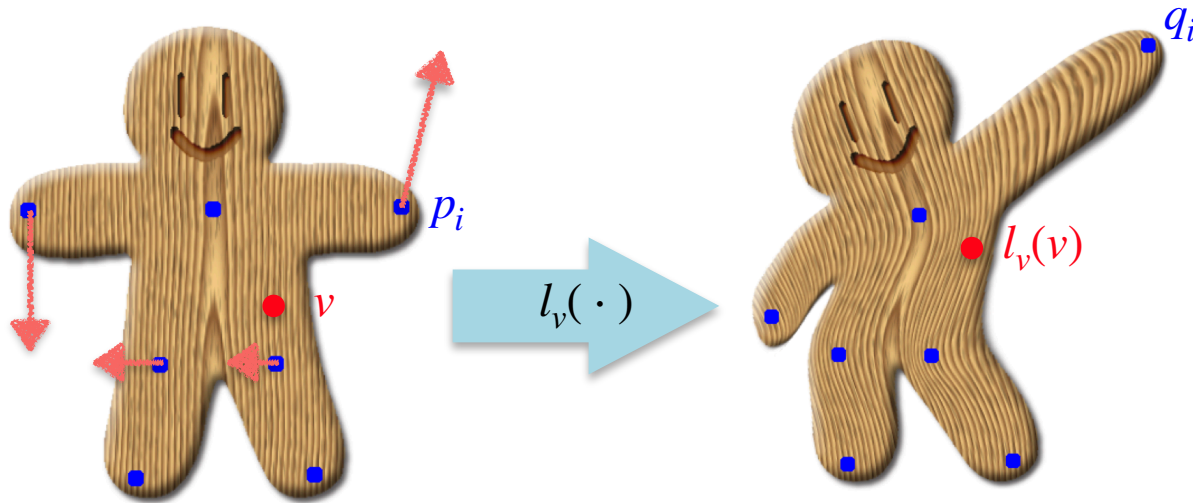
## Today

- New exercise, deadline in **3** weeks – 09.04.2022 at 11:59pm

## This week

- Grades for Exercise 1 should be available by the end of this week

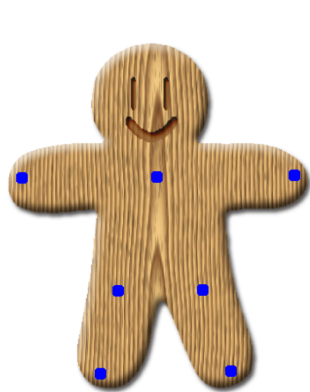
# Task 1: Image Warping



$$\sum_i w_i |l_v(p_i) - q_i|^2, w_i = \frac{1}{|p_i - v|^{2\alpha}} \quad l_v(x) = xM + T$$

Backward/Reverse Mapping: [https://www.cs.princeton.edu/courses/archive/spr11/cos426/notes/cos426\\_s11\\_lecture03\\_warping.pdf](https://www.cs.princeton.edu/courses/archive/spr11/cos426/notes/cos426_s11_lecture03_warping.pdf)

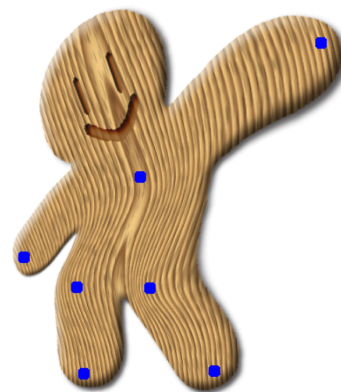
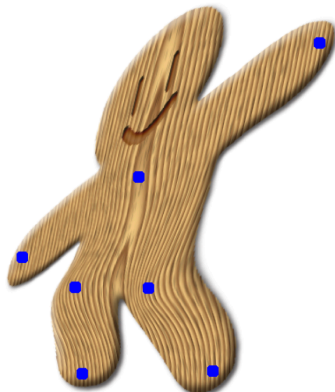
# Task 1: Image Warping



$$l_v(x) = xM + T$$

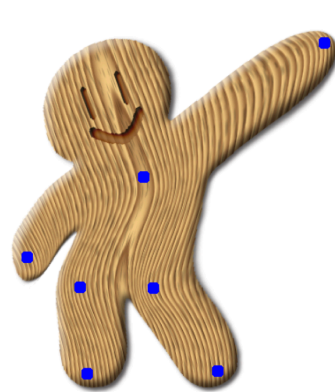
## Affine

- Translation
- Rotation
- Non uniform scaling
- Shear



## Similarity

- Translation
- Rotation
- Uniform scaling



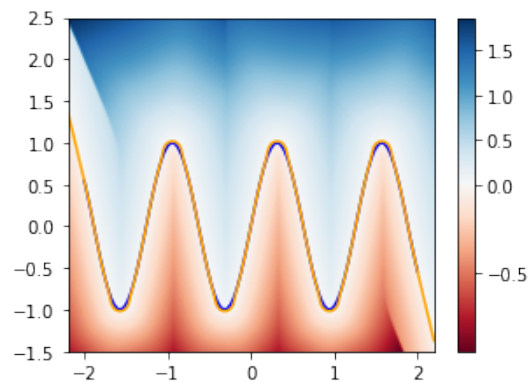
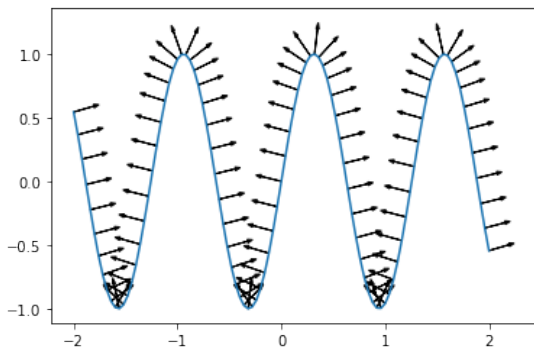
## Rigid

- Translation
- Rotation

Image Deformation Using Moving Least Squares, Schaefer et al., SIGGRAPH 2006  
<https://people.engr.tamu.edu/schaefer/research/mls.pdf>

# Task 2: Curve Reconstruction

- Sampled data for 3 curves is provided (x, y and normals)
- Derive implicit function (locally fit constant polynomial)
- Evaluate implicit function on regular grid using point sets



# Task 2: Curve Reconstruction

- Implicit MLS surfaces (Lecture 2)

Minimized for  
each point you  
evaluate on


$$\operatorname{argmin}_{c_0} \sum_i \phi_i(\|x - x_i\|)(c_0 - f_i(x))^2$$

$$\phi_i(\|x - x_i\|) = e^{-\frac{\|x - x_i\|^2}{\sigma^2}} \quad f_i(x) = n_i^T(x - x_i)$$

# Task 3: Image Compression

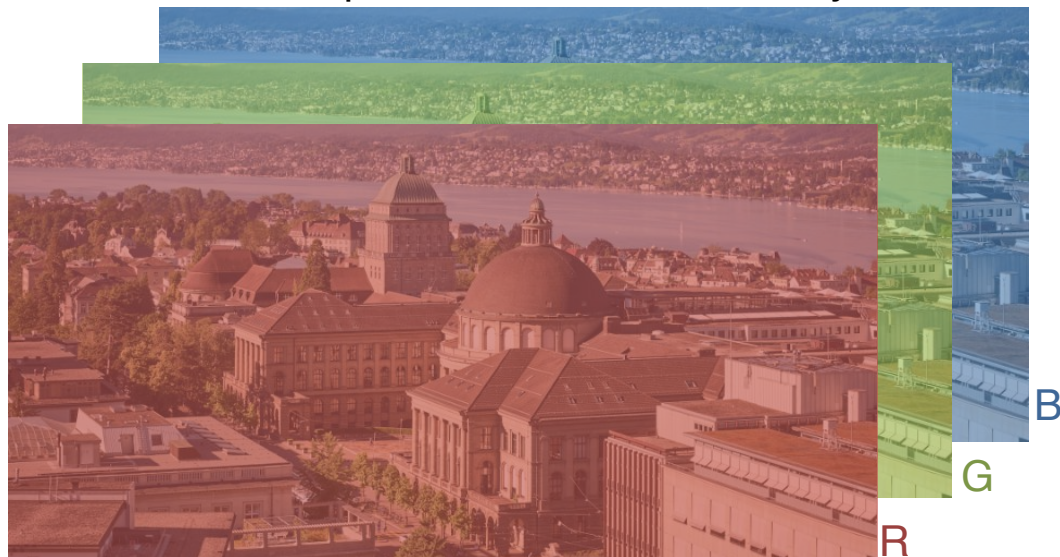
- Single image provided



$$3 \times H \times W$$

# Task 3: Image Compression

- Single image provided
- Use PCA to compress image along rows
- Problem: Cannot perform PCA with 3D array



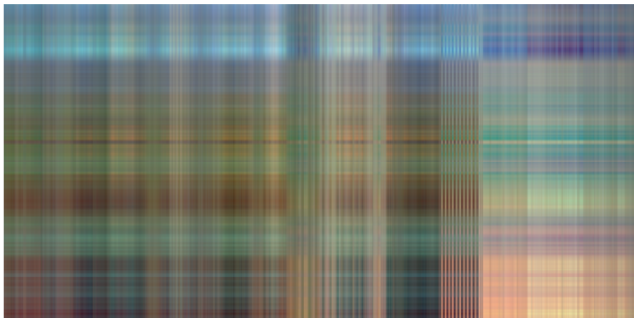
$H \times W(\text{num\_data} \times \text{data\_dimension})$

Solution:

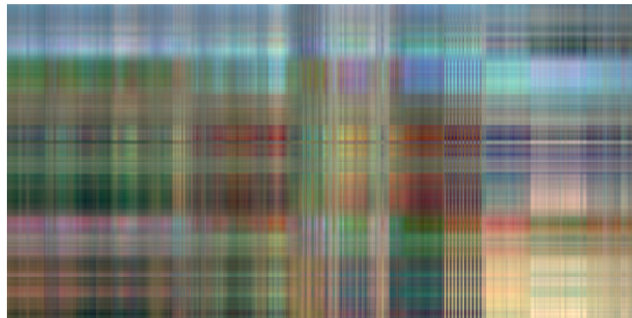
1. Split the original image in to R, G and B channels
2. Perform PCA on each channel
3. Merge the compressed channels to get compressed image

# Task 3: Image Compression

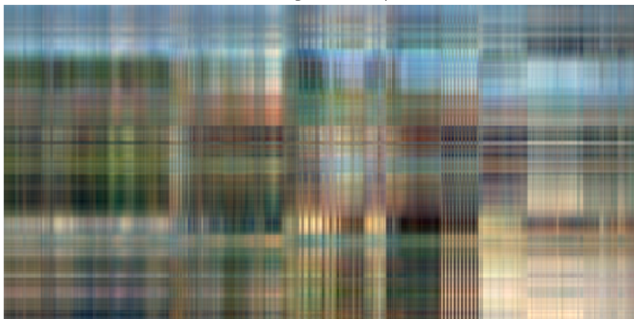
Reduced image on 1 components



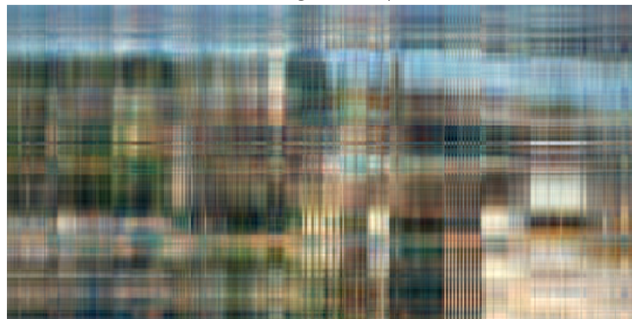
Reduced image on 2 components



Reduced image on 3 components



Reduced image on 5 components



# End of Handout 3

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## Questions?